

Multiple Choice (20 marks)

1. Which environmental factor would most likely prevent an orange tree from surviving in the desert?
 - (a) the lack of moisture
 - (b) the high levels of heat
 - (c) the extra amount of wind
 - (d) the low levels of sunlight
2. An example of tool use is
 - (a) mankind's closes relative using a stick to get ants out of a hole
 - (b) a person throwing a hammer into a pool
 - (c) a bird building a nest
 - (d) someone looking at screwdrivers in a store
3. When a wave moves from shallow water to deep water, the
 - (a) frequency decreases, the wavelength increases, and the speed increases
 - (b) frequency does not change, the wavelength increases, and the speed increases
 - (c) frequency does not change, the wavelength decreases, and the speed decreases
 - (d) frequency does not change, the wavelength increases, and the speed decreases
4. Which of the following was NOT a factor in the extinction of the Neandertals?
 - (a) They were specialized for an environment that suddenly changed.
 - (b) Their tool technology was less advanced.
 - (c) They were unsuited to the hotter climates of the sub-Saharan savanna.
 - (d) The Neandertal body type required greater amounts of calories to sustain.
5. Who is likely to receive its coloration and hairy body from its parents?
 - (a) Peter the Parakeet
 - (b) Valentina the Vole
 - (c) Terry the Tuna
 - (d) Greta the Grouper
6. A plant produces tiny plants around the edges of its leaves. When these tiny plants fall to the ground, they take root and become new plants. This process is...

- (a) sexual reproduction
 - (b) asexual reproduction
 - (c) evolution
 - (d) extinction
7. The ability to solve a new computer game based on logical puzzles probably depends on which kind of intelligence?
- (a) fluid intelligence
 - (b) crystallized intelligence
 - (c) aptitude intelligence
 - (d) achievement intelligence
8. Which of the following refers to the benefit of having an emotional release to reduce aggressive tendencies?
- (a) James-Lange theory of emotion
 - (b) Counterconditioning
 - (c) Catharsis hypothesis
 - (d) Transference
9. The brown fur of the arctic hare turns white in winter. How does this color change most likely help the arctic hare?
- (a) It helps the animal save water.
 - (b) It helps the animal hide from predators.
 - (c) It helps keep the animal cool.
 - (d) It helps protect the animal from disease.
10. In the absence of air resistance a ball of mass m is tossed upward to reach a height of 20 m. At the 10-m position half way up the net force on the ball is
- (a) $2mg$
 - (b) mg
 - (c) $mg/2$
 - (d) $mg/4$

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The *ad nauseam* fallacy is' is 'reasoning that a conclusion is true because it's been claimed so often'.
2. True or False: The correct answer to 'Which of the following information about publications does the Audit Bureau of Circulation NOT provide?' is 'Traffic'.
3. True or False: The correct answer to 'Which action will result in a product with new chemical properties?' is 'popping popcorn'.
4. True or False: The correct answer to 'condensation is a stage in what cycle process?' is 'solar'.
5. True or False: The correct answer to 'The classification of some organisms has changed. Which new process is used to reclassify organisms?' is 'Structures are now examined at the molecular level.'.
6. True or False: The correct answer to "'*Ad crumenam*" is a specific kind of' is 'False sign'.
7. True or False: The correct answer to 'Which culture hosted the above celebration?' is 'the Ancestral Puebloans of the southwestern desert'.
8. True or False: The correct answer to 'Which of the following is the investigation of the hedonic consumption of products?' is 'Social interaction perspective.'.
9. True or False: The correct answer to 'Let A be a finite set with m elements, and let B be a finite set with n elements. The number of distinct functions mapping A into B is' is ' $n!/(m!(n - m)!)$ '.
10. True or False: The correct answer to 'The reaction that releases energy for use by the cell occurs in which site?' is 'Golgi apparatus'.

End of Paper